The Toy Workshop Theft

Whodunnit? · Similarity

Name: _____

At Gearhart's Toy Workshop, the inventor was about to show the world his greatest invention: **Tiny Titan**, a walking, talking toy robot. But the cloth was lifted and the stand was bare!

Six people were in the workshop, each in a different spot, and there are six ways to sneak a toy robot out the door. You are the workshop detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Mr. Gearhart Toy Inventor	☐ Design Studio	☐ Packed in a toy box
☐ Ms. Pom-Pom Designer	☐ Paint Room	☐ Hidden in a teddy bear
☐ Ben Tran Apprentice	☐ Assembly Line	☐ Rolled out on a skateboard
☐ Dr. Doodle Toy Tester	☐ Testing Lab	☐ Dressed as a doll
☐ Mrs. Wick Packager	☐ Packing Dock	☐ Shipped in a gift box
☐ Mr. Spoole Delivery Driver	☐ Teddy Corner	☐ Stashed in a shopping bag

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Toy Workshop Theft · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Toy Workshop Theft

△ABC ~ △DEF. AB = 4, BC = 6 and DE = 6. Find EF.

A. 9	Mr. Gearhart was sketching designs in the Design Studio. The late-night guard saw him.
B. 10	Every teddy bear on the shelf was squeezed and shaken. Nothing hard inside any of them.
C. 4	Ben Tran was sharpening pencils in the Design Studio. Mr. Gearhart said he stayed all afternoon.
D. 8	Mr. Gearhart was checking the gears of a toy dog in the Testing Lab. The tester signed his visit.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Toy Workshop Theft

Solve the proportion:

$$\frac{x}{15} = \frac{4}{10}$$

A. 60	The teddy bears were all sewn shut with tight factory stitches. None had been torn open.
B. 9	Ms. Pom-Pom was testing a rainbow robot in the Testing Lab. The lab log lists her name.
C. 37.5	Ben Tran was mixing paint in the Paint Room. Blue splotches cover his apron.
D. 6	Every toy box was opened and emptied. No Tiny Titan.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Toy Workshop Theft

A triangle with a side of 6 is enlarged to a similar triangle with the matching side 9. Find the scale factor.

A. 3	Mrs. Wick was wrapping a prize in the Testing Lab. The testers gave her a round of applause.
B. $\frac{6}{15}$	The teddy bears are too small and floppy to hold a robot. Nobody could hide Tiny Titan in one.
C. $\frac{3}{2}$	Ms. Pom-Pom was painting a rocking horse in the Paint Room. Wet paint all over her sleeves proves it.
D. $\frac{2}{3}$	Ben Tran was oiling gears on the Assembly Line. The foreman watched him the whole shift.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Toy Workshop Theft

Two similar polygons have a scale factor of 2 : 5. What is the ratio of their perimeters?

A. $\frac{2}{5}$	The skateboard in the hall has a flat wheel. It has not rolled in days.
B. $\frac{8}{125}$	Mr. Spoole was dropping off a package at the Testing Lab. The sign-in sheet has his name.
C. $\frac{5}{2}$	Teddy Corner’s bears were counted at closing. Every bear was accounted for and untouched.
D. $\frac{4}{25}$	Ben Tran was loading pallets at the Packing Dock. The dock camera caught him every minute.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Toy Workshop Theft

Two similar polygons have a scale factor of 2 : 5. What is the ratio of their areas?

A. $\frac{4}{25}$	Dr. Doodle was testing wind-up cars on the Assembly Line with three other testers.
B. $\frac{25}{4}$	The teddy bear with the torn ear was checked. Only stuffing inside.
C. $\frac{8}{125}$	Dr. Doodle was timing a toy race in the Testing Lab. The stopwatch still shows the time.
D. $\frac{2}{5}$	Ben Tran was fluffing pillows in Teddy Corner. The kids giggled as he worked.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Toy Workshop Theft

A 6-ft person casts a 4-ft shadow. A tree casts a 14-ft shadow at the same time. How tall is the tree?

A. $\frac{28}{3}$ ft	Mr. Gearhart was tightening bolts on a toy crane in the Testing Lab. His wrench is still there.
B. 21 ft	The dolls on the shelf are standing in neat rows, each one a normal-sized doll.
C. 18 ft	The bears were scanned at the gate. The pictures show only fluff.
D. 10 ft	Ben Tran was drawing blueprints in the Design Studio. The ink was still wet.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

Rectangle A is 3 by 5. Rectangle B is similar with a short side of 9. Find the long side of Rectangle B.

A. 5.4	Ben Tran was washing brushes in the Paint Room. The sink is still full of colorful water.
B. 11	Ms. Pom-Pom was handing out stickers in the Testing Lab. The kids all remember her.
C. 27	The teddy bear on the night shelf has a tag dated yesterday. It was never opened.
D. 15	Mrs. Wick was taping boxes at the Packing Dock, timed by the dock camera.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

On a map, 1 inch represents 25 miles. Two cities are 3.5 inches apart on the map. How many miles apart are they?

A. 875 mi	Every bear’s tummy zipper was found zipped and locked. No one had opened a single one.
B. 87.5 mi	The gift boxes were opened and counted. All held the gifts they should.
C. 28.5 mi	Mrs. Wick was delivering boxes to the Testing Lab. The delivery slip is stamped.
D. 0.14 mi	Ben Tran was fixing a conveyor belt on the Assembly Line with two other workers.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Toy Workshop Theft

In △ABC, DE ∥ BC with D on AB and E on AC. AD = 4, DB = 6 and AE = 5. Find EC.

A. 9	The teddy bears were lined up and counted at the door. Not one was missing or lumpy.
B. $\frac{10}{3}$	Mr. Spoole was waiting outside the Testing Lab for a signature. The clerk kept him company.
C. 7.5	Mr. Spoole was reading stories to kids in Teddy Corner for the whole hour.
D. 12.5	Ben Tran was sweeping the Packing Dock. The pile of packing peanuts is proof.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Toy Workshop Theft

Two similar figures have areas 18 and 50. Find the scale factor of the smaller to the larger.

A. $\frac{9}{25}$	Ben Tran was reading a picture book in Teddy Corner. The kids sat all around him.
B. $\frac{3}{5}$	The shopping bags were checked at the door. Only lunches and umbrellas.
C. $\frac{5}{3}$	Dr. Doodle was drop-testing a toy truck in the Testing Lab. The dents prove it.
D. $\frac{25}{9}$	The big teddy bear by the door was checked. Its stuffing was fluffy and fresh.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.